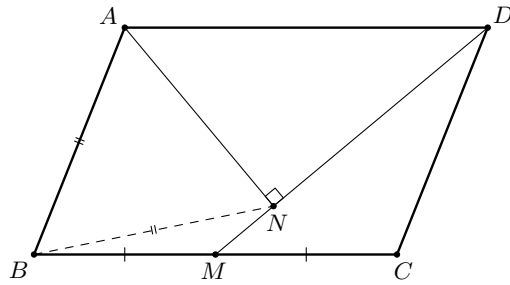


HK IMO Training 2018-2019

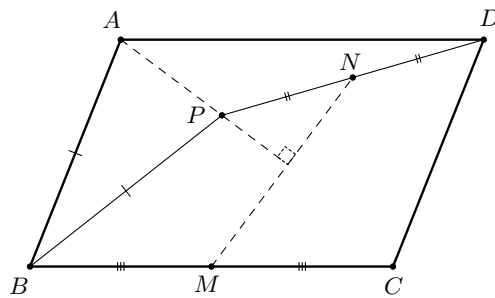
Parallelograms

1 Geometry in Figures: Parallelograms

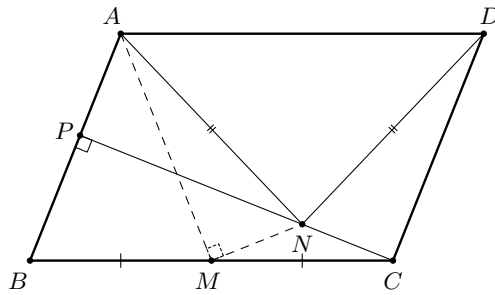
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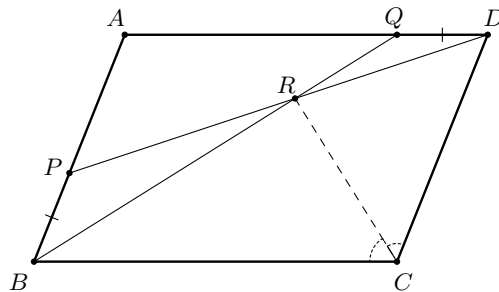
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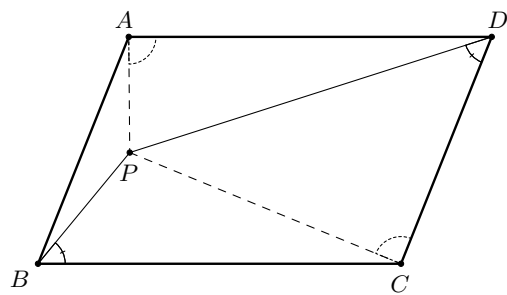
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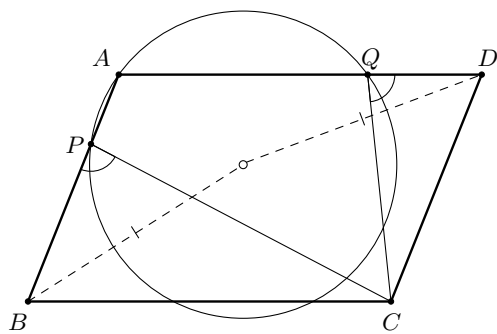
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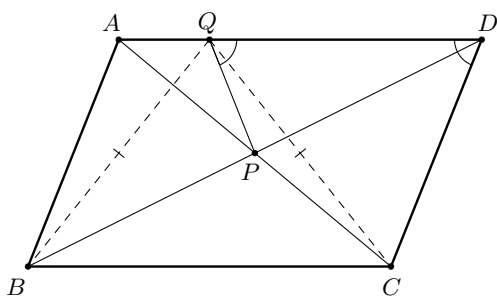
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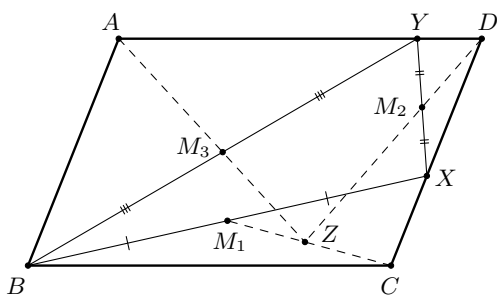
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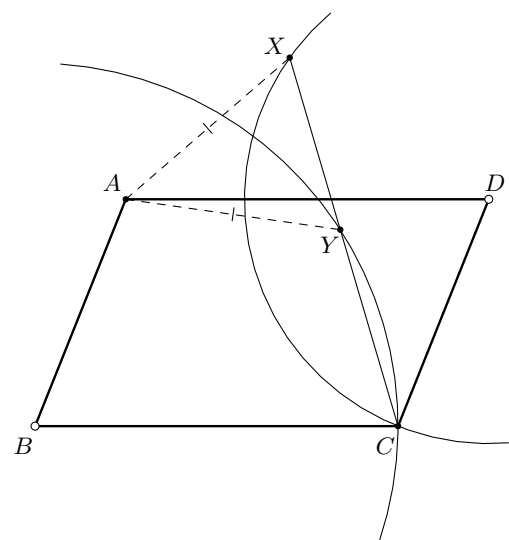
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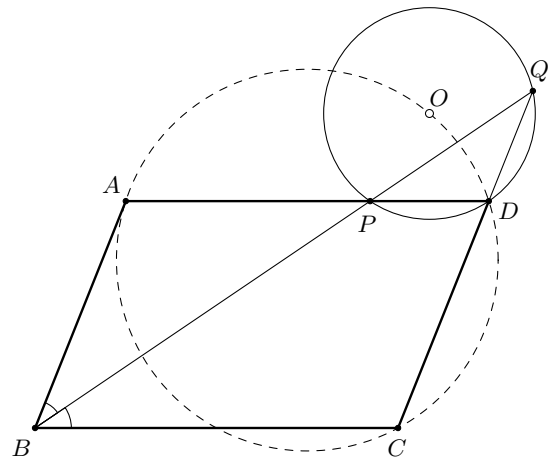
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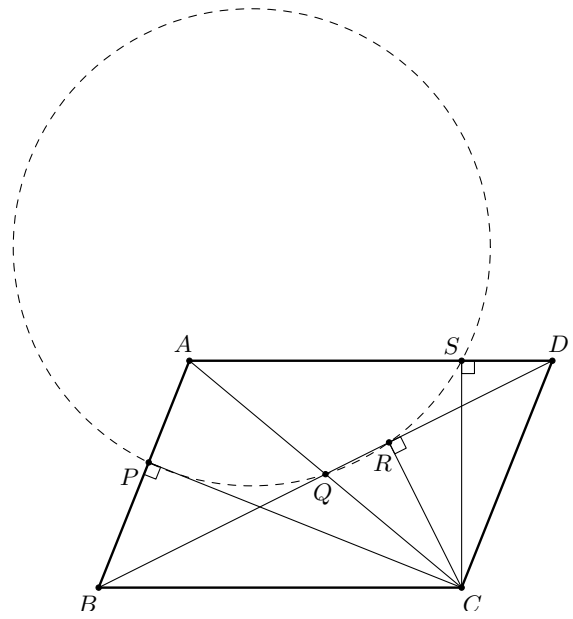
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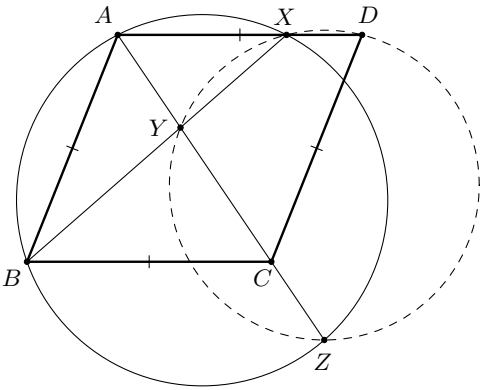
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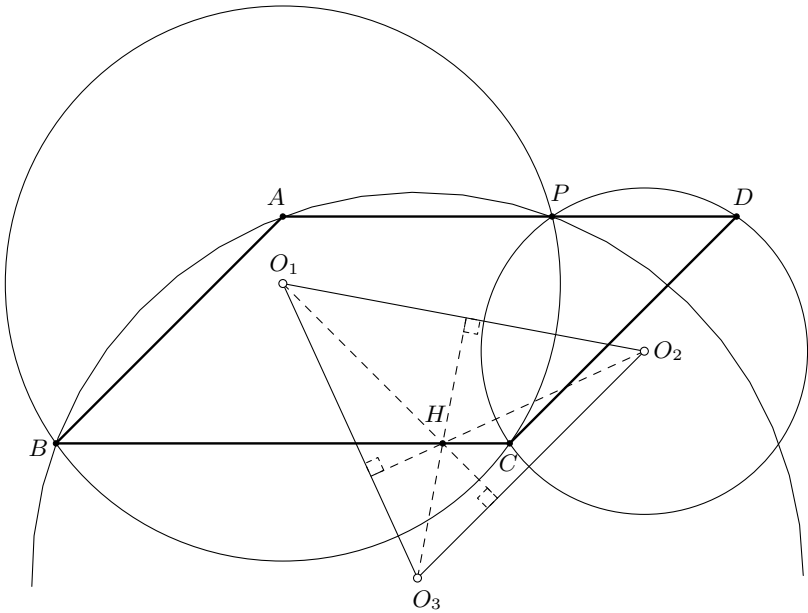
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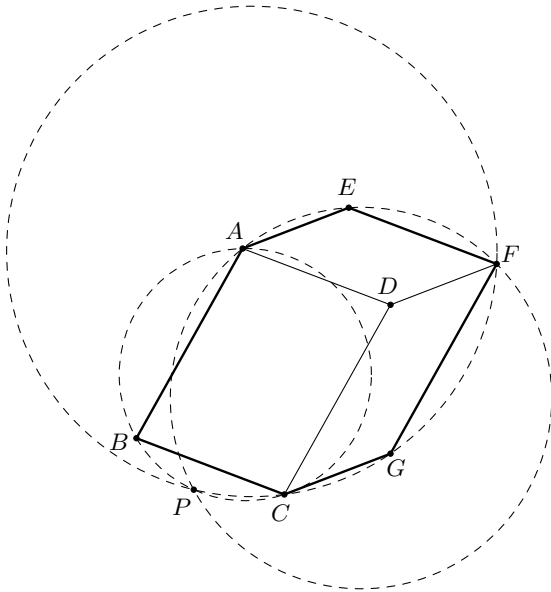
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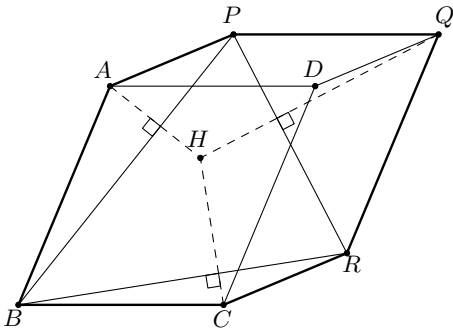
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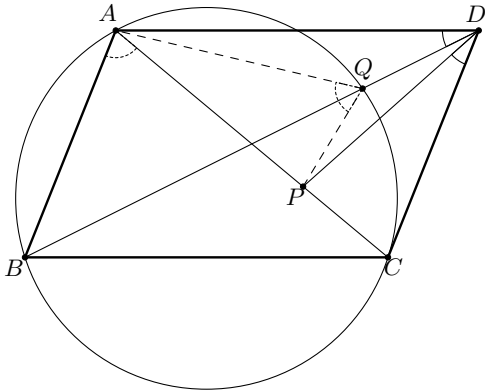
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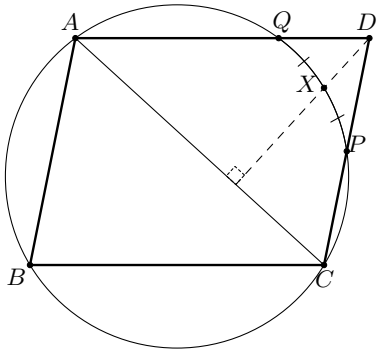
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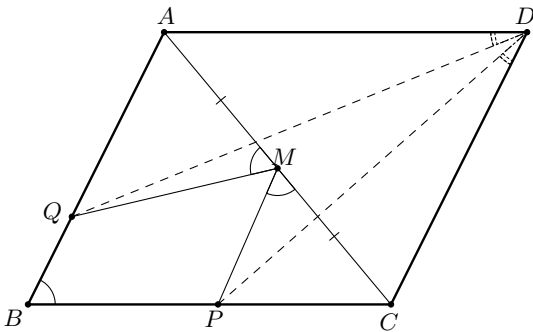
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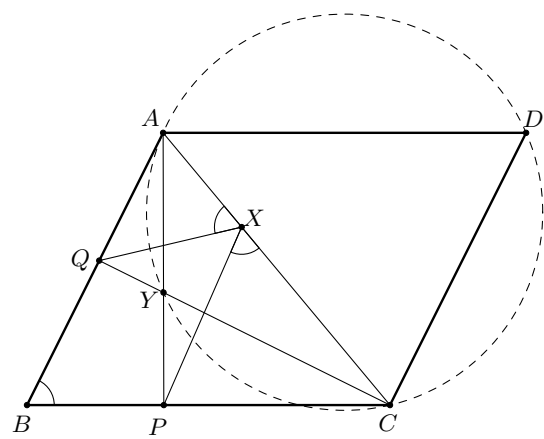
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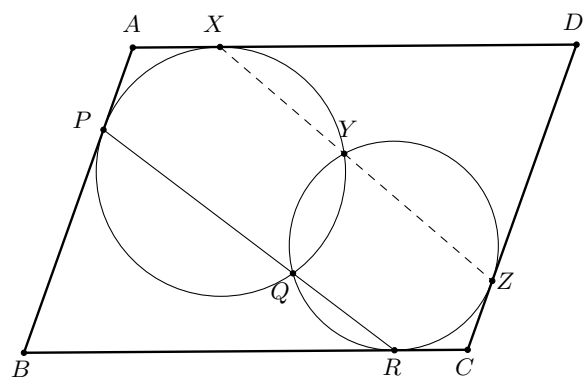
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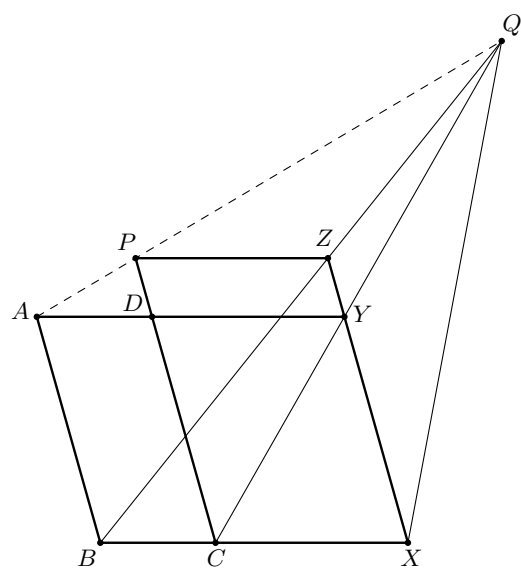
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5.1.20



5.1.21



2 Miscellaneous Problems on Parallelograms

Problem 1 (Geometry Revisited 1.9 no.3). The point P lies inside triangle ABC so that $\angle ABP = \angle PCA$. The point Q is such that $PBQC$ is a parallelogram. Prove that $\angle QAB = \angle CAP$.

Problem 2 (Turkey Team Selection Test for JBMO 2018). A point E lies inside a parallelogram $ABCD$ such that $\angle BAE = \angle BCE$. Show that the circumcentres of triangles ABE , BCE , CDE and DAB are concyclic.

Problem 3 (Turkey 1996, no. 2). In a parallelogram $ABCD$ with $\angle A < 90^\circ$, the circle with diameter AC meets the lines CB and CD again at E and F , respectively, and the tangent to this circle at A meets BD at P . Show that P, F, E are collinear.

Problem 4 (APMO 2010 Problem 1). Let ABC be a triangle with $\angle BAC \neq 90^\circ$. Let O be the circumcentre of $\triangle ABC$ and let Γ be the circumcircle of $\triangle BOC$. Suppose that Γ intersects the line segment AB at P different from B ; and the line segment AC at Q different from C . Let ON be a diameter of the circle Γ . Prove that the quadrilateral $APNQ$ is a parallelogram.

Problem 5 (Turkey 1996, no. 2). In a parallelogram $ABCD$ with $\angle A < 90^\circ$, the circle with diameter AC meets the lines CB and CD again at E and F , respectively, and the tangent to this circle at A meets BD at P . Show that P, F, E are collinear.

Problem 6 (Belarussia 2018 Category D no. 6). Let $ABCD$ be a parallelogram with $BC \parallel AD$ and $BC = 2AB$. The internal angle bisector of $\angle ABC$ intersects the side AD at point K and the diagonal AC at point L . The internal angle bisector of $\angle ADC$ intersects the extension of AB beyond B at point M . The line ML intersects the AD at F . Find the ratio $AF : AD$.

Problem 7 (2012 G2). Let $ABCD$ be a cyclic quadrilateral whose diagonals AC and BD meet at E . The extensions of the sides AD and BC beyond A and B meet at F . Let G be the point such that $ECGD$ is a parallelogram, and let H be the image of E under reflection in AD . Prove that D, H, F, G are concyclic.

Problem 8 (2007 G4). Consider five points A, B, C, D, E such that $ABCD$ is a parallelogram and $BCED$ is a cyclic quadrilateral. Let ℓ be a line passing through A , and let ℓ intersect segment DC and line BC at points F and G , respectively. Suppose that $EF = EG = EC$. Prove that ℓ is the bisector of angle DAB .

Problem 9 (Russia 1996, no.18). In an isosceles triangle ABC with $AB = BC$, one draws the angle bisector CD . The perpendicular to CD through the centre of the circumcircle of ABC intersects BC at E . The parallel to CD through E meets AB at F . Show that $BE = FD$.

Problem 10 (Russia 2000 no.24). The point K is inside a parallelogram $ABCD$ such that midpoint Q of AD is equidistant from K and C and such that the midpoint of CD is equidistant from K and A . Let N be the midpoint of BK . Prove that $\angle NAK = \angle NCK$.

Problem 11 (Ibero-America 1997, no.5). Let AE and BF be altitudes, and H the orthocentre, of an acute triangle ABC . The reflection of AE across the interior angle bisector of A meets the reflection of BF across the interior angle bisector of B in a point O . The lines AE and AO meet the circumcircle of ABC again at M and N , respectively. Let P, R, S be the intersection of BC with HN , BC with OM , HR with OP , respectively. Show that $AHSO$ is a parallelogram.

Problem 12 (MEMO 2018 T-5). Let ABC be an acute-angled triangle with $AB < AC$, and let D be the foot of its altitude from A . Points B' and C' lie on the rays AB and AC , respectively, so that points B', C' and D are collinear and points B, C, B' and C' lie on one circle with centre O . Prove that if M is the midpoint of BC and H is the orthocentre of $\triangle ABC$, then $DHMO$ is a parallelogram.

Problem 13 (MEMO 2010 T-6). Let A, B, C, D, E be points such that $ABCD$ is a cyclic quadrilateral and $ABDE$ is a parallelogram. The diagonals AC and BD intersect at S and the rays AB and DC intersect at F . Prove that $\angle AFS = \angle ECD$.

Problem 14 (IGO 2018, Intermediate Level no.5). Suppose that $ABCD$ is a parallelogram such that $\angle DAC = 90^\circ$. Let H be the foot of perpendicular from A to DC , also let P be a point along the line AC such that the line PD is tangent to the circumcircle of the triangle ABD . Prove that $\angle PBA = \angle DBH$.